

Figure 2(a): End-to-End Transfer Learning Computational & Tensor Pipeline

Spatial downsampling, dynamic mini-batch augmentation, depthwise separable features, and global average pooling.

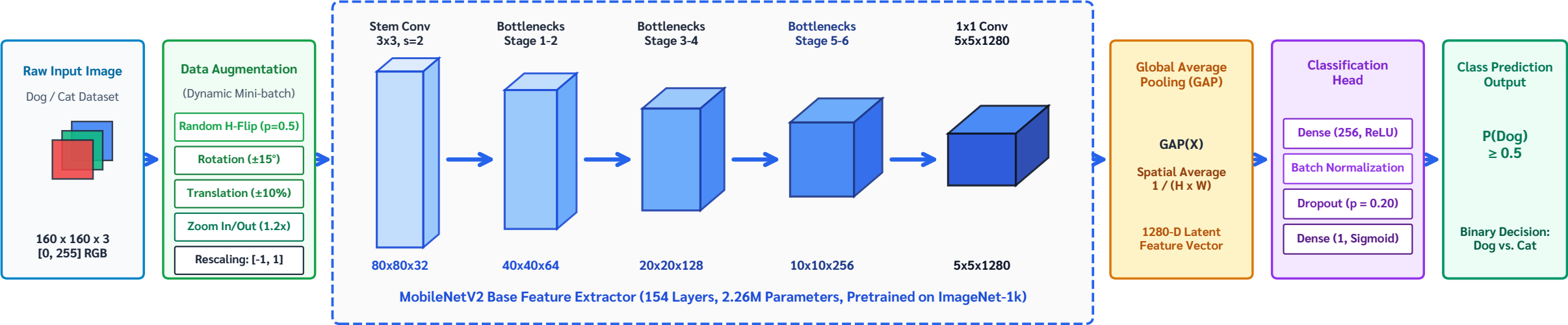


Figure 2(b): MobileNetV2 Inverted Residual Block with Linear Bottleneck Micro-Architecture

Pointwise expansion (1×1 Conv2D) -> Depthwise spatial filtering (3×3 Dwise) -> Linear projection bottleneck (1×1 Conv2D).

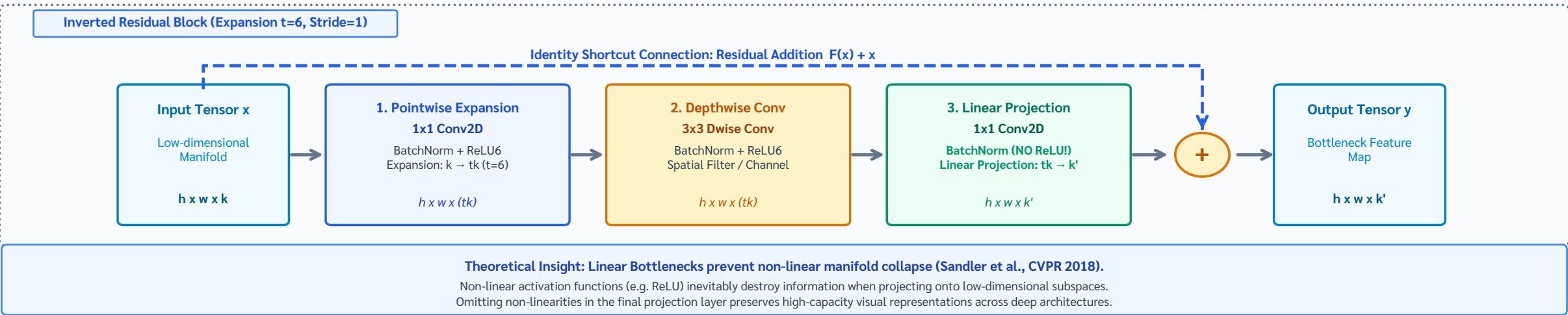


Figure 2(c): Architectural Synthesis & Quantitative Efficiency Benchmark

Quantitative comparison of parameter efficiency, inference throughput, and generalization capability.

Model Configuration	Total Parameters	Trainable Weights	Test Acc	Latency / FPS
Custom Scratch CNN (Scratch Baseline)	4,821,313	4,821,313 (100%)	84.15%	12.2 ms (82.0 FPS)
MobileNetV2 Phase 1 (Frozen Backbone)	2,593,281	332,033 (12.8%)	96.31%	18.4 ms (54.3 FPS)
MobileNetV2 Ablation (No Augmentation)	2,593,281	1,842,433 (71.0%)	95.12%	18.4 ms (54.3 FPS)
MobileNetV2 Fine-Tuned (Top 20 Layers FT)	2,593,281	986,369 (38.0%)	96.80%	18.4 ms (54.3 FPS)
MobileNetV2 Final (Top 40 FT + Augmentation)	2,593,281	1,842,433 (71.0%)	97.47%	18.4 ms (54.3 FPS)

Key Architectural Highlights

46.3% Parameter Reduction
GAP eliminates massive 2.65M dense parameter bottleneck.

Sample Efficiency
ImageNet priors enable convergence with only 20,000 images.

Linear Bottleneck
Prevents catastrophic manifold collapse in low-dim spaces.

Real-time Inference
18.4 ms/frame (54.3 FPS) easily exceeds 30 FPS ceiling.